Alireza **Amiri**

MECHATRONICS ENGINEER · CONTROL SYSTEMS SPECIALIST

Karaj, Iran

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“Eager to innovate, learn, and solve problems with determination and teamwork.”

Objective

Eager to find and develop solutions for upcoming problems by combining innovation and hard work. Experienced in diverse fields, yet passionate about learning new skills and exploring new challenges.

Education

K. N. Toosi University of Technology — *Tehran, Iran*

MASTER OF SCIENCE IN MECHATRONICS — *Sep. 2023 – Present*

Shahed University — *Tehran, Iran*

BACHELOR OF SCIENCE IN ELECTRICAL ENGINEERING (CONTROL ORIENTATION) — *Jul. 2019 – Jul. 2023*

- GPA: 16.03/20
- Thesis: Design and Implementation of a PID-based Control System for Achieving Stable Magnetic Levitation

Experience

MAPNA Generator Engineering & Manufacturing Co. — *Karaj, Iran*

MAINTENANCE AND REPAIR ENGINEERING INTERN — *Jun. 2022 – Sep. 2022*

- Gained insights into industrial large-scale factories and basic performance of industrial machinery.
- Learned fault diagnosis procedures and cost-efficient problem-solving techniques.
- Experimented with industrial control systems, including PLC and LOGO systems.

Haftonim Publication — *Remote*

TRANSLATOR — *Mar. 2021 – Jul. 2021*

- Translated a scientific book, ”Archimedes and the Door of Science,” for teenage readers.
- Developed skills in literary translation and adapted text to Persian.

Self-Employed — *Remote*

NFT MARKETING — *Jan. 2022 – Jul. 2022*

- Launched and marketed ”Callipen,” a calligraphy artwork brand.
- Achieved international sales and developed cross-cultural communication skills.

Selected Projects

Magnetic Levitation

- Developed an Arduino-based PID controller to achieve stable magnetic levitation.
- Experimented with PID and H-infinity control methods using MATLAB and Simulink.
- Designed and optimized an electrical circuit to support system stability.

Speech Emotion Detection

- Trained a machine learning model to classify emotions and gender from Persian speech data.
- Achieved 60% accuracy in the best-case scenario.

ECG Signal Monitoring

- Designed an ESP32-based system for recording ECG signals and heart rates.
- Collaborated on biomedical device development, learning about medical sensors.

Skills

Programming Languages Python, Arduino, Verilog, C++

Engineering Tools MATLAB, Simulink, Proteus, ISE

Soft Skills Problem Solving, Time Management, Logical Thinking, Teamwork, Communication

Languages

Persian Native

English Fluent

Arabic Conversational